

SATHWIK BUSIREDDY

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EDUCATION

GITAM (DEEMED TO BE UNIVERSITY), School of Technology, *Hyderabad, TS, IND*
Bachelor's in Computer Science and Engineering.

April 2023

UNIVERSITY OF GEORGIA, Franklin College of Arts and Sciences, *Athens, GA, USA*
Master's in Computer Science

May 2025

SKILLS SUMMARY

- **Programming Languages:** C, Java, Python, R.
- **Database:** MySQL, MongoDB.
- **Data Warehousing:** Snowflake, Amazon Redshift.
- **Data Analysis & Visualization:** Excel, Tableau, Power BI, Matplotlib.
- **Data Engineering Tools:** PySpark, Apache Airflow, Kafka.
- **Concepts:** Data Structures and Algorithms, Object Oriented Programming, Machine Learning, Quantitative Analysis, Risk Management.
- **Project Management Methodology:** Agile, Scrum
- **Cloud:** AWS, Azure, Google cloud platform.

EXPERIENCE

University of Georgia

2024-2025

Research Assistant

- Designed and trained deep learning models using TensorFlow to classify food inspection outcomes with over 22,000 records, boosting prediction accuracy by more than 14% compared to baseline models.
- Integrated LLMs to generate human-readable explanations for model decisions and recommend risk mitigation strategies, improving interpretability.
- Applied transfer learning and hyperparameter tuning, reducing overfitting and increasing generalization across real-world datasets by 9%.

University of Georgia Center for Continuing Education & Hotel

2023-2025

Student Assistant Manager

- Directed daily operations for a team of 150+ staff, increasing productivity KPIs by 30% over three semesters through effective delegation and mentorship.
- Used Jira to manage tasks across departments, improving task completion rate from 87% to 94%.
- Collaborated with leadership to implement strategic workflow changes that reduced operational bottlenecks and improved guest satisfaction metrics.

Dhyanahitha Pvt.Ltd.

2022-2022

Software Intern:

- Created backend APIs and optimized database queries for a tweet classification platform, reducing response time by 16%.
- Trained supervised ML models on Twitter data, achieving 87% accuracy in multi-class sentiment classification.
- Collaborated in Agile teams to deliver full-stack features, gaining proficiency in end-to-end application development.

PERSONAL PROJECTS

Predictive Analysis model for Food Inspection

2024-2025

Python, TensorFlow, Scikit-learn, Snowflake, RAG, Sequential Neural Network, React.js, Django REST Framework.

- Developed a neural network using TensorFlow that achieved over 87% accuracy in predicting food inspection outcomes on a real-world dataset of 22,000+ records
- Utilized Snowflake for storing and querying preprocessed inspection data, enabling efficient large-scale data handling and SQL-based analytics.
- Implemented a Retrieval-Augmented Generation (RAG) pipeline with Gemini's LLM to generate natural language explanations for predictions and compared interpretability versus the neural model alone
- Designed and deployed a RESTful Django API backed by a React.js interface to serve predictions and explanations in real-time for end users

Time Series Forecasting and Trend Analysis

2024-2024

Python, Pandas, NumPy, Matplotlib, Seaborn, Statsmodels (ARIMA), SQL, AWS EC2, Apache Airflow, Power BI

- Built a robust time series model using ARIMA in Python to forecast trends, achieving 92% R² accuracy on validation data
- Engineered features and cleaned 10,000+ time-indexed records using SQL and Pandas for seasonal pattern analysis.
- Designed the pipeline for potential scheduling via Apache Airflow and deployed initial visualization using Flask and Power BI.

Intrusion Detection of Imbalanced Network Traffic

2022-2023

Python, Scikit-learn, Random Forest, SVM, CNN, LSTM, Matplotlib

- Led a team to develop a machine learning-based intrusion detection system that achieved 11% higher accuracy in identifying anomalies in imbalanced datasets.
- Combined ensemble and deep learning to detect threats in real time from over 100,000 simulated network packets.
- Visualized detection patterns and performance metrics using Matplotlib and Seaborn to interpret model effectiveness.